A super learner for heterogeneous net treatment effects[a]

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Abstract

We introduce a super learner for estimating heterogeneous net treatment effects under unit-varying outcome and cost effects. Our approach is designed for optimal assignment of a binary treatment that induces a cost–benefit trade-off. The net effect and its underlying outcome and cost components are characterized by unknown functional complexity, which our ensemble explores in a data-driven manner.

Directly targeting the net effect performs well when the estimand is simpler than the outcome and cost effects individually. In contrast, separately learning both effects and subsequently aggregating them into the net effect is advantageous when the components are structurally dissimilar or when their aggregation yields a target estimand that is more complex than the components themselves. A hybrid learning strategy succeeds in intermediate settings. Our ensemble nests all approaches and selects the winner by minimizing empirical risk.

In a simulation study, we consider multiple scenarios in which each individual approach dominates the others and show that the ensemble further improves precision across most settings. We additionally present an empirical application using data from a large nonprofit organization, where we analyze the net effect of a fundraising campaign aimed at increasing pledge payments while mitigating donor attrition.

Keywords: Causal inference, cost-benefit analysis, treatment rule, super learner
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1 Introduction

Combining causal estimands into one target parameter of interest also introduces flexibility to translate the target parameter into an estimate. One economically relevant example is the heterogeneous net benefit of a binary treatment, where the treatment variable affects both the outcomes and costs of a unit based on covariates. The net effect directly relies on the underlying outcome and cost effects, which may either be modeled jointly or separately conditional on covariates and other nuisance parameters. The modeling choice provides a broader scope for the implicit assumptions an estimator makes about the functional complexity of the target, which is called inductive bias.

Suppose a decision maker wishes to assign the treatment in a way that maximizes the net benefit across the treated population. Cost efficiency requires the marginal benefit from the treatment, the outcome effect, to at least cover the arising marginal costs, the cost effect. If costs are constant across units, it suffices to estimate Conditional Average Treatment effect (CATE) based on the outcome variable and use the constant cost as the cut-off for treatment assignment. However, if costs vary at the individual level, both personalized components must be taken into account, which increases the complexity of the target and, ultimately, the functional complexity one wishes to learn. The key challenge is that the appropriate level of aggregation, whether to target the net effect directly or to estimate its components separately, depends on this unknown structure.

We propose super learning for complex causal estimands as a solution to inductive bias and introduce an ensemble method for estimating the Conditional Net Average Treatment Effect (CNATE). The main contribution of this paper is our proposed method designed to derive unit-level effects in a data-driven manner. We showcase how the underlying complexity of the response surfaces and the resulting outcome and cost effects shape the complexity of the estimand, which motivates the use of an ensemble. Our approach can further be used for approximating the target population where cost efficiency is satisfied.

We illustrate our method with an empirical example using data from a non-governmental organisation (NGO) that aims to increase pledge donations and targets their donors by fundraising communication. This may have the desired effect of higher payments, but may also serve as a reminder of the active payment the donor plans to cancel and leads to drop out. We show that a targeting mechanism would increase welfare and leverage the targeting operator characteristic to investigate heterogeneity across the population (Yadlowsky, Fleming, Shah, Brunskill, & Wager, 2025).

Our paper relates to three strands of literature. First, it discusses inductive bias in the context of heterogeneous treatment effect estimation. Inductive bias has been discussed by Künzel, Sekhon, Bickel, and Yu (2019), who use different meta-learner algorithms for estimating the CATE. Hahn, Murray, and Carvalho (2020) describe the problem of regularization-induced confounding and bias in CATE estimation, while Caron, Baio, and Manolopoulou (2022) introduce a comprehensive framework comparing different meta-learner strategies. Curth and van der Schaar (2021b) show that standard indirect CATE estimators implicitly bias toward treatment effect heterogeneity, and propose strategies that instead encode an inductive bias favoring shared structure between the potential outcomes. Most recently, Liang, van der Laan, and Alaa (2025) propose a hybrid meta-learner that internally regularizes between direct and indirect learning strategies. Our work extends this discussion by allowing for varying unit-level costs measured in a separate variable.

Second, we use super learning (Dudoit & van der Laan, 2005; Sinisi, Polley, Petersen, Rhee, & van der Laan, 2007) to combine different learning strategies for CNATE estimation instead of relying on a single procedure. Third, we contribute to the literature on cost-benefit analysis in economics. Zhao and Harinen (2020) model net value optimization using modified meta-learners including the outcome and cost effect. Cost-effectiveness analysis in medicine jointly estimates treatment effects on both costs and health outcomes (Baio, 2012; Bonander & Svensson, 2021; Esser et al., 2025; Gabrio, Baio, & Manca, 2019).

Third, our approach relates to the literature on targeted policy making. Sun, Munro, Kalashnov, Du, and Wager (2024) derive a priority score for targeted assignment using the cost-benefit ratio. In fundraising, recent work derives optimal targeting rules that maximize donations while accounting for campaign costs (Athey, Byambadalai, Cersosimo, Koutout, & Nath, 2024; Breman, 2011; Cagala, Glogowsky, Rincke, & Strittmatter, 2021; Esterzon, Lemmens, & Van den Bergh, 2023), but costs are usually treated as fixed and known. In our empirical application, we analyze how to optimally target donors when both solicitation costs and donation benefits are uncertain and possibly correlated.

This article is organized as follows. Section 2 presents a motivating example that illustrates the role of inductive bias in CNATE estimation. Section 3 introduces the framework and notation. Section 4 formalizes the inductive bias argument. Section 5 introduces the super learner. We analyze finite-sample properties in Section 6 and provide an application in Section 7. Section 8 concludes.

2 Motivating example

Before introducing the formal framework, we illustrate the core challenge of CNATE estimation through a stylized example. Let Y_i be the outcome and C_i the cost of unit i

under a binary treatment $W_i \in \{0,1\}$. The net effect for unit i is $\psi(X_i) = \tau_Y(X_i) - \tau_C(X_i)$, where τ_Y and τ_C denote the outcome and cost effects. The key question is whether the level of aggregation at which one estimates ψ amplifies or simplifies its functional complexity relative to its components. We consider two extreme cases.

We model the potential outcomes using conditional response surfaces $\mu^*_{Y(0)}(X_i)$ and $\mu^*_{Y(1)}(X_i)$, and the potential costs using $\mu^*_{C(0)}(X_i)$ and $\mu^*_{C(1)}(X_i)$. Sine and step functions characterize the response surfaces.

Scenario 1: Constant CNATE. With a single covariate $X_i = x$ and zero cost in the absence of treatment ($\mu^*_{C(0)}(X_i) = 0$), let

$$\mu^*_{Y(0)}(x) = 1 - 0.5\sin(3x), \quad \mu^*_{Y(1)}(x) = 1.5 + 0.5\sin(3x), \quad \mu^*_{C(1)}(x) = \sin(3x). \tag{1}$$

Then $\tau_Y(x)$ and $\tau_C(x)$ are the same function shifted by the constant $\psi(x) = 0.5$, which precisely equals the net effect.[1] As Figure 1 (A) shows, transforming the outcomes (orange) provides a better fit of the true CNATE (black) compared to separately estimating the outcome and cost effects and subsequently differencing (blue). The reason is that aggregating the response surfaces simplifies the target parameter.

Scenario 2: Heterogeneous CNATE. Let $X_i = (X_1, X_2)$ with $X_i = x$ and $x = (x_1, x_2)$ consist of two independent covariates, and set

$$\begin{aligned} \mu^*_{Y(0)}(X_i) &= \sin(x_1), & \mu^*_{Y(1)}(X_i) &= 1.5 + 2\sin(x_1)\,\mathbf{1}\{x_1 > -1\}, \\ \mu^*_{C(0)}(X_i) &= 0.1 + \sin(2x_2), & \mu^*_{C(1)}(X_i) &= 3\,\mathbf{1}\{x_2 > 1\}. \end{aligned} \tag{2}$$

[1] An even simpler setting with a constant net effect can be constructed by setting the outcome and cost effects to constants, too.

Here, the outcome- and cost-effects are driven by different covariates and combine into a highly complex net effect. In this setting, it is advantageous to optimize the fit of all components in isolation. Thus, separately estimating the outcome and cost effects and subsequent differencing (blue) yields the best fit for the true CNATE function, as shown in Figure 1 (B), where we stratify by low vs. high values of X_1. Note that in Scenario 1, the transformed-outcome learner achieves a lower RMSE against the true CNATE than the separate learner (0.048 vs. 0.072), whereas in Scenario 2 the ranking reverses sharply (0.396 vs. 0.185).

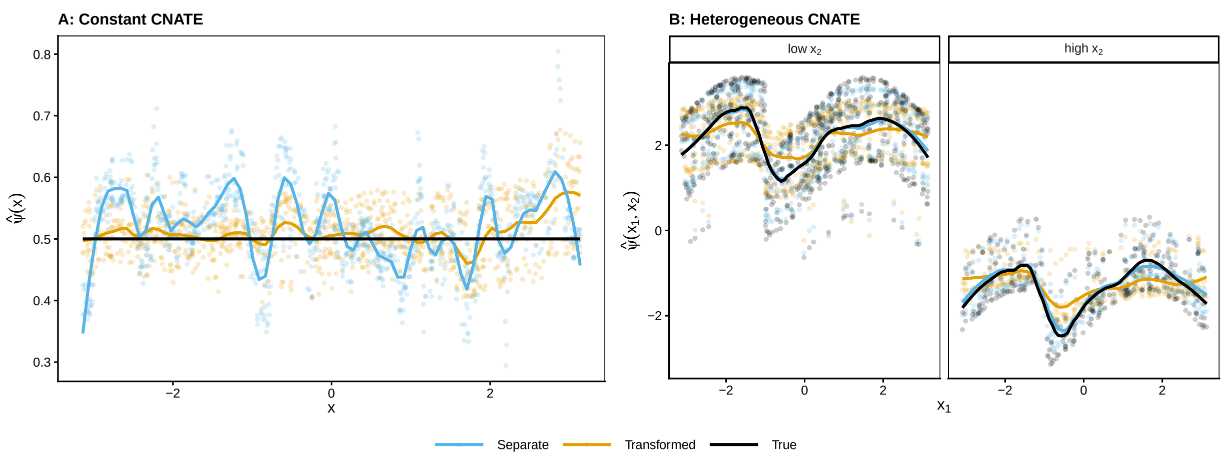

Figure 1: *Motivating example: constant vs. heterogeneous CNATE. The left panel shows Scenario 1 where the CNATE is constant; the right panel shows Scenario 2 where it varies with x_1 and x_2. Lines show LOESS-smoothed point predictions from the transformed-outcome learner (orange) and the separate-difference learner (blue) against the true CNATE (black).*

As we do not know the underlying complexity of the target, we propose optimizing over an ensemble that includes all learning approaches and a weighted average thereof. This motivates our use of the super learner in Section 5. We formalize the framework in the following section.

3 Framework

Data and setup. We observe n independent and identically distributed tuples $O_i = (X_i, Y_i, C_i, W_i)$, with units $i = 1, \dots, n$ drawn from a distribution P, where $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$ is a covariate vector and $W_i \in \{0, 1\}$ is a binary treatment indicator. Moreover, $Y_i \in \mathbb{R}$ is the outcome of unit i, and $C_i \in \mathbb{R}$ is the cost associated with treatment. Throughout, capital letters denote random variables, lowercase letters denote their realizations. For any function f, we write $f(x)$ when defining the function at a fixed point $x \in \mathcal{X}$, and $f(X_i)$ when evaluating it at the covariates of unit i. We follow the notation in Nie and Wager (2020) and use $*$-superscripts to denote unknown population quantities and hats for their estimates. A function without star or hat denotes a generic candidate being optimized over.

Estimand. Adopting the potential outcomes framework (Neyman, 1923; Rubin, 1974), let $Y_i(w)$ and $C_i(w)$ denote the potential outcomes and potential costs for unit i under treatment $w \in \{0, 1\}$. We define the conditional net average treatment effect (CNATE) as our estimand by

$$\psi^*(x) \coloneqq \mathbb{E}\big[\{Y_i(1) - Y_i(0)\} - \{C_i(1) - C_i(0)\} \;\big|\; X_i = x\big], \tag{3}$$

where the expectation is over the common true data-generating distribution P^* of any unit i. Following the notation in M. J. van der Laan and Rose (2011, 2018), we let $\mathcal{M}$ be the statistical model, i.e., a collection of candidate distributions P for O_i. The estimand in (3) is a mapping $\Psi\colon \mathcal{M} \to \mathcal{F}$, where $\mathcal{F}$ is a suitable function space on $\mathcal{X}$. For any $P \in \mathcal{M}$, we write $\psi_P(x) = \Psi(P)(x)$ for the CNATE implied by P. Under the true data-generating distribution P^*, this yields $\psi^*(x) \equiv \psi_{P^*}(x) = \Psi(P^*)(x)$.

Identification. Identification of $\psi^*(x)$ from the observed data distribution relies on the following assumptions, formally stated in Assumption 1. First, treatment assignment is unconfounded, meaning that the treatment status is independent of potential outcomes and costs, conditional on observed covariates X_i. Second, the propensity score $e^*(x) = \Pr(W_i = 1 \mid X_i = x)$ is bounded away from zero and one, ensuring that for each value of X_i, comparable units exist in both the treatment and control group. Third, there is no interference between units, so that one individual's treatment assignment does not affect another individual's outcomes or costs, and the observed outcome and cost for each unit equal the potential outcome and cost corresponding to the treatment actually received.

Assumption 1. *Identifying causal assumptions.*

(a) Unconfoundedness: $\{Y_i(0), Y_i(1), C_i(0), C_i(1)\} \perp\!\!\!\perp W_i \mid X_i$.

(b) Overlap (common support): There exists $\varepsilon > 0$ *such that* $\varepsilon < e^*(x) < 1 - \varepsilon$ *for all* $x \in \mathcal{X}$, *where* $e^*(x) = \Pr(W_i = 1 \mid X_i = x)$ *is the propensity score and* $\mathcal{X}$ *denotes the support of* X_i.

(c) Stable unit treatment value assumption (SUTVA): We have consistency of observed and potential outcomes and costs by

$$Y_i = W_i\,Y_i(1) + (1 - W_i)\,Y_i(0),$$

$$C_i = W_i\,C_i(1) + (1 - W_i)\,C_i(0).$$

Moreover, the potential outcomes $\{Y_i(w), C_i(w)\}_{w \in \{0,1\}}$ *depend only on the treatment assignment* W_i *of unit* i *itself, and are not affected by the treatment assignments* W_j *of any other unit* $j \neq i$.

Under Assumption 1, the CNATE in (3) is identified from the observed data O_i; see Appendix A.1. Due to the linearity of the expectation in (3), there exist several estimation strategies that are asymptotically equivalent but differ in finite samples (Liang et al., 2025).

Estimation strategies. We estimate $\psi^*(x)$ by empirical loss minimization of a doubly robust loss function. Throughout, we adopt three variants of the R-loss introduced in Nie and Wager (2020), summarized in Table 1.[2] We denote $\mu_Y^*(x), \mu_C^*(x)$, and $e^*(x)$ as nuisance parameters, where $\mu_Y^*(x) = \mathbb{E}[Y_i \mid X_i = x]$ is the conditional mean outcome, $\mu_C^*(x) = \mathbb{E}[C_i \mid X_i = x]$ is the conditional mean cost, and $e^*(x)$ the propensity score from Assumption 1(b).[3]

The transformed R-learner uses $Z_i = Y_i - C_i$ and $\mu_Z^*(x) = \mathbb{E}[Z_i \mid X_i = x]$ and directly targets $\psi^*(x)$ by minimizing $R_{\text{tf}}(\psi)$ to obtain estimates of $\hat{\psi}_{\text{tf}}(x)$. The joint R-learner also targets $\psi^*(x)$ directly but includes nuisances $\mu_Y^*(x)$ and $\mu_C^*(x)$ separately when minimizing $R_{\text{joint}}(\psi)$ to obtain estimates $\hat{\psi}_{\text{joint}}(x)$. The separate R-learner first minimizes two independent loss functions, $R_Y(\tau_Y)$ and $R_C(\tau_C)$, by individually targeting the CATE of outcomes and costs,

$$\tau_Y^*(x) = \mathbb{E}\big[\{Y_i(1) - Y_i(0)\} \bigm| X_i = x\big], \tag{4}$$

$$\tau_C^*(x) = \mathbb{E}\big[\{C_i(1) - C_i(0)\} \bigm| X_i = x\big], \tag{5}$$

before the final estimator takes the difference $\hat{\psi}_{\text{sep}}(x) = \hat{\tau}_Y(x) - \hat{\tau}_C(x)$.[4] Section 4 contrasts

[2]Our approach could similarly be implemented using a different doubly robust loss, such as the AIPW-loss. We use the R-loss because it estimates the CNATE in the region with overlap and therefore performs well with observational data, avoiding instability when propensity scores lie close to the boundaries (Foster & Syrgkanis, 2023; Morzywolek, Decruyenaere, & Vansteelandt, 2024).

[3]For a binary treatment, $e^*(x) = \Pr(W_i = 1 \mid X_i = x) = \mathbb{E}[W_i \mid X_i = x]$.

[4]A similar estimation strategy can be found in Bonander and Svensson (2021).

the corresponding estimators based on the different R-learner loss functions in Table 1 in more detail. To fix ideas for the corresponding cross-fitted estimators, we present the CNATE estimator for the transformed R-learner in Section 5 in Equation (7). The joint and separate R-learner estimators, $\hat{\psi}_{\text{joint}}(x)$ and $\hat{\psi}_{\text{sep}}(x) = \hat{\tau}_Y(x) - \hat{\tau}_C(x)$, are defined analogously by replacing R_{tf} with R_{joint} or with the pair (R_Y, R_C) from Table 1.

Table 1: *R-learner loss functions overview.*

Strategy	*R-loss*
I. Transformed	$R_{\text{tf}}(\psi) = \mathbb{E}\left[\left((Z_i - \mu^*_Z(X_i)) - (W_i - e^*(X_i))\,\psi(X_i)\right)^2\right]$
II. Joint	$R_{\text{joint}}(\psi) = \mathbb{E}\left[\left((Y_i - \mu^*_Y(X_i)) - (C_i - \mu^*_C(X_i)) - (W_i - e^*(X_i))\,\psi(X_i)\right)^2\right]$
III. Separate	$R_Y(\tau_Y) = \mathbb{E}\left[\left((Y_i - \mu^*_Y(X_i)) - (W_i - e^*(X_i))\,\tau_Y(X_i)\right)^2\right]$
	$R_C(\tau_C) = \mathbb{E}\left[\left((C_i - \mu^*_C(X_i)) - (W_i - e^*(X_i))\,\tau_C(X_i)\right)^2\right]$

Notes: Summary of R-learner losses that target the CNATE in (3). At the population level with true nuisances, R_{tf} and R_{joint} coincide, since $Z_i = Y_i - C_i$ and $\mu^*_Z(X_i) = \mu^*_Y(X_i) - \mu^*_C(X_i)$ by linearity of conditional expectations. The two strategies diverge only at the empirical plug-in stage: R_{tf} uses a single nuisance estimate $\hat{\mu}_Z$, whereas R_{joint} substitutes the pair $(\hat{\mu}_Y, \hat{\mu}_C)$ separately. Strategy III goes one step further by also separating the regression stage that targets the treatment effect itself: each component CATE of either outcome or cost is estimated under its own loss and complexity penalty before differencing.

4 Inductive bias

The motivating example in Section 2 illustrates that the appropriate estimation strategy depends on the unknown functional complexity of the CNATE and its components. We now formalize this observation. The key question is whether the level of aggregation in Table 1 amplifies or cancels the complexity of the estimand in (3). By utilizing the approaches I.–III., we implicitly impose assumptions on how to learn $\psi^*(x)$. This set of assumptions about the underlying complexity can be considered as inductive bias.

Inductive bias in CATE estimation. In CATE estimation based on outcome Y_i only, inductive bias refers to the assumptions an estimator makes about the relationship between conditional response surfaces $\mu^*_{Y(0)}(X_i)$ and $\mu^*_{Y(1)}(X_i)$, whose contrast $\tau^*_Y(x)$ is the estimand but is never observed for any unit (Holland, 1986). Two key axes determine this bias: the degree of structure shared between potential outcomes, ranging from full pooling to full separation (Curth & van der Schaar, 2021b; Hahn et al., 2020; Künzel et al., 2019), and whether one regularizes the potential outcomes indirectly or $\tau^*_Y(x)$ directly. This matters since direct targeting can exploit simpler effect structure (Curth & van der Schaar, 2021a; Kennedy, 2023).

Extension to the CNATE. The discussion on inductive bias for $\tau^*_Y(x)$ directly connects to our analysis of the CNATE and may amplify the issue due to further uncertainty about the cost effect $\tau^*_C(x)$ and the nuisance component $\mu^*_C(x)$. If the assumptions encoded in the estimator about the complexity of $\psi^*(x)$ and the underlying relationships between $\tau^*_Y(x)$, $\tau^*_C(x)$, $\mu^*_Y(x)$, $\mu^*_C(x)$ deviate from the truth, the estimator may perform poorly in finite samples.

Aggregation level and estimation error. The three strategies in Table 1 span a spectrum of aggregation levels. At one extreme, the transformed R-learner is the most parsimonious approach and collapses the problem immediately by using a single composite outcome Z_i. However, estimating the CNATE in one step forces a single nuisance model to capture the joint structure of outcomes and costs. At the other extreme, the separate R-learner treats outcomes and costs as entirely distinct causal inference problems, estimating each treatment effect with its own tailored nuisance functions before combining them by subtraction. While this offers maximal flexibility, it may accumulate estimation error from two independent learning problems. The joint R-learner occupies a middle ground: it retains separate nuisance models to respect the potentially different signal structures of outcomes and costs while still targeting the net effect directly in a single loss function.

Motivation for the ensemble. In any given application, the functional form of the nuisance functions, the heterogeneity of the causal effects, and the way they combine into the CNATE are all unknown. Each of these unknowns can influence how well an estimator performs in finite samples. Instead of committing to a single estimation strategy upfront, we combine all three learners through a super learner (Dudoit & van der Laan, 2005; Sinisi et al., 2007), which uses cross-validation to determine the optimal weighting of candidates based on their empirical performance. Section 5 embeds the estimation of the CNATE with the R-learners in Table 1 into the super learner framework.

In any given application, the functional form of the nuisance functions, the heterogeneity of the causal effects, and the way they combine into the CNATE are all unknown, and each of these unknowns can materially influence which strategy performs best in finite samples. Section 5 embeds the three R-learners into an ensemble that resolves this uncertainty in a data-driven manner.

5 Super learner

Cross-fitting setup. To avoid overfitting, all nuisance functions, base learners, and stacking weights are obtained via V-fold cross-fitting. Let $\mathcal{D} = \{O_1, \ldots, O_n\}$ denote the full sample, partitioned into V disjoint folds with validation index sets $\mathrm{Val}(1), \ldots, \mathrm{Val}(V)$, and let $v(i)$ denote the fold containing observation i. Throughout, a superscript $(-v)$ on a fitted function means training on the complement $\{i : i \notin \mathrm{Val}(v)\}$ and the trained function may then be evaluated at any x.

Nuisance step. For each $v \in \{1, \ldots, V\}$, the nuisances $\hat{\mu}_Z^{(-v)}, \hat{\mu}_Y^{(-v)}, \hat{\mu}_C^{(-v)}, \hat{e}^{(-v)}$ are fit on the complement of fold v. Evaluating each at its own held-out fold and pooling yields the out-of-fold (OOF) residuals

$$\tilde{Z}_i = Z_i - \hat{\mu}_Z^{(-v(i))}(X_i), \quad \tilde{Y}_i = Y_i - \hat{\mu}_Y^{(-v(i))}(X_i), \quad \tilde{C}_i = C_i - \hat{\mu}_C^{(-v(i))}(X_i), \quad \tilde{W}_i = W_i - \hat{e}^{(-v(i))}(X_i), \tag{6}$$

defined for every $i = 1, \ldots, n$. By construction, the nuisance model producing the residual at observation i never saw i during training.

Base learner step. For each v, the fold-v transformed R-learner is trained on the complement using the OOF residuals from (6):

$$\hat{\psi}_{\mathrm{tf}}^{(-v)} = \arg\min_{\psi} \frac{1}{n - n_v} \sum_{i \notin \mathrm{Val}(v)} \left(\tilde{Z}_i - \tilde{W}_i \, \psi(X_i)\right)^2 + \Lambda_n(\psi), \tag{7}$$

where $\Lambda_n(\psi)$ is a regularizer on the complexity of $\psi(\cdot)$ in the spirit of Nie and Wager (2020). The joint and separate R-learners $\hat{\psi}_{\mathrm{joint}}^{(-v)}$ and $\hat{\psi}_{\mathrm{sep}}^{(-v)} = \hat{\tau}_Y^{(-v)} - \hat{\tau}_C^{(-v)}$ are defined analogously by replacing the transformed loss with R_{joint} or with the pair (R_Y, R_C) from Table 1. Evaluating each fold-v learner at its held-out fold and pooling yields the OOF base-learner predictions

$$\hat{\psi}_k^{\mathrm{oof}}(X_i) \;:=\; \hat{\psi}_k^{(-v(i))}(X_i), \qquad i = 1, \ldots, n, \quad k \in \{\mathrm{tf}, \mathrm{joint}, \mathrm{sep}\}, \tag{8}$$

each of which is out-of-sample with respect to its own observation.

Candidate library. We define a library of K candidate estimators $\hat{\psi}_k$, indexed by $k \in \{1, \ldots, K\}$. In our specific case, $K = 4$ with $k \in \{\text{tf}, \text{joint}, \text{sep}, \text{stack}\}$. The first three candidates correspond to the transformed, joint, and separate R-learners defined above. The fourth candidate is a continuous super learner that combines the three base learners through a non-negative weighted average,

$$\hat{\psi}_{\text{stack}}(x) = \hat{\alpha}_1 \, \hat{\psi}_{\text{tf}}(x) + \hat{\alpha}_2 \, \hat{\psi}_{\text{joint}}(x) + \hat{\alpha}_3 \, \hat{\psi}_{\text{sep}}(x), \qquad \hat{\alpha}_1, \hat{\alpha}_2, \hat{\alpha}_3 \geq 0. \tag{9}$$

The stacking weights are themselves cross-fitted and the fold-v weights solve a ridge-regularised constrained least-squares problem on the complement. The stacking step is an optimization across held-out base-learner predictions and nuisance residuals. Hence, the fold-v stacked prediction at its held-out fold gives, after pooling, $\hat{\psi}^{\text{oof}}_{\text{stack}}(X_i)$ for all $i = 1, \ldots, n$. We explain the stacking procedure in Appendix A.4 in more detail.

Empirical risk and winner selection. All four candidates are scored against a common cross-validated empirical R-risk computed on the OOF predictions,

$$\hat{R}_{\text{CV}}(\hat{\psi}_k) = \frac{1}{n} \sum_{i=1}^{n} \big(\tilde{Z}_i - \tilde{W}_i \, \hat{\psi}^{\text{oof}}_k(X_i) \big)^2. \tag{10}$$

Equivalently, (10) can be written as $\frac{1}{V} \sum_{v=1}^{V} \frac{1}{n_v} \sum_{i \in \text{Val}(v)} (\cdot)$ when folds are equal-sized. Note that (10) coincides with the transformed-loss objective in (7) without the regularizer Λ_n, scored against any candidate $\hat{\psi}_k$ rather than minimized over ψ. The discrete super learner selects the candidate with the smallest cross-validated empirical risk,

$$\hat{k}_n = \arg \min_{k \in \{\text{tf},\text{joint},\text{sep},\text{stack}\}} \hat{R}_{\text{CV}}(\hat{\psi}_k), \tag{11}$$

and the final CNATE estimate is $\hat{\psi}^*(x) = \hat{\psi}_{\hat{k}_n}(x)$. The discrete winner is preferred over directly using the continuous stack as the final estimator, since it inherits the theoretical guarantees of the best candidate in the library, whereas the convex combination may not always identify the global minimum of the empirical risk (Dudoit & van der Laan, 2005). The full procedure is summarised in Algorithm 1.

Algorithm 1 Super learner for the CNATE

Require: Training data $\{(X_i, Y_i, C_i, W_i)\}_{i=1}^n$, number of folds V, candidate library $\{\hat{\psi}_k\}_{k=1}^K$.

```
Partition {1, …, n} into V folds Val(1), …, Val(V).
for v = 1, …, V do
    Fit nuisances μ̂_Y^(−v), μ̂_C^(−v), μ̂_Z^(−v), ê^(−v) on {i : i ∉ Val(v)}.
end for
Form OOF residuals Z̃_i, Ỹ_i, C̃_i, W̃_i for all i via (6).
for v = 1, …, V do
    Fit base learners ψ̂_k^(−v), k ∈ {tf, joint, sep}, on {i : i ∉ Val(v)} using Z̃_i, Ỹ_i, C̃_i, W̃_i.
end for
Form OOF base-learner predictions ψ̂_k^oof(X_i) for all i and k via (8).
Set ψ̂_tjs^oof(X_i) = (ψ̂_tf^oof(X_i), ψ̂_joint^oof(X_i), ψ̂_sep^oof(X_i))^⊤.
for v = 1, …, V do
    Estimate stacking weights α̂^(−v) on {i : i ∉ Val(v)} via (A.19).
    Set ψ̂_stack^oof(X_i) = α̂^(−v)⊤ ψ̂_tjs^oof(X_i) for i ∈ Val(v)
end for
Compute R̂_CV(ψ̂_k) for all k via (10).
Select winner k̂_n via (11).
return ψ̂*(x) = ψ̂_{k̂_n}(x).
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Theoretical guarantees. We derive high-probability regret bounds for the super learner. We assume the following regularity conditions:

Assumption 2 (Regularity conditions)**.**

(a) ***Bounded predictions.*** *There exists $M < \infty$ such that $\|\hat{\psi}_k^{(-v)}\|_\infty \leq M$ for all $k \in \{1, \dots, K\}$ and all $v \in \{1, \dots, V\}$.*

(b) ***Sub-Gaussian noise.*** *The residuals $Y_i - \mu_Y^*(X_i)$, $C_i - \mu_C^*(X_i)$, $Z_i - \mu_Z^*(X_i)$, and $W_i - e^*(X_i)$ are sub-Gaussian conditional on X_i.*

(c) ***Nuisance convergence rates.*** *All nuisance estimators converge at rate $n^{-1/4}$ in $L^2(P)$ on held-out folds: for each v,*

$$\|\hat{\mu}_Z^{(-v)} - \mu_Z^*\|_{L^2} \vee \|\hat{\mu}_Y^{(-v)} - \mu_Y^*\|_{L^2} \vee \|\hat{\mu}_C^{(-v)} - \mu_C^*\|_{L^2} \vee \|\hat{e}^{(-v)} - e^*\|_{L^2} = o_p(n^{-1/4}). \quad (12)$$

Assumption 2(c) is the standard cross-fitting rate condition of Chernozhukov et al. (2018). It is satisfied by a broad class of nonparametric estimators, including random forests and regularised regression, whenever the nuisance functions belong to a sufficiently smooth function class (L. van der Laan, 2026). Assumption 2(a) can be relaxed to a moment condition at the cost of additional notation.

Theorem 1 (Oracle inequality)**.** *Suppose Assumptions 1 and 2 hold. Let $R(\psi) = \mathbb{E}\Big[(\tilde{Z}_i^* - \tilde{W}_i^*\,\psi(X_i))^2\Big]$ denote the population R-risk at true nuisances, where $\tilde{Z}_i^* = Z_i - \mu_Z^*(X_i)$ and $\tilde{W}_i^* = W_i - e^*(X_i)$. For each cross-fitted candidate $\hat{\psi}_k$, define the fold-averaged population risk*

$$\bar{R}(\hat{\psi}_k) \;:=\; \frac{1}{V}\sum_{v=1}^{V} R(\hat{\psi}_k^{(-v)}),$$

where $R(\cdot)$ acts on each $\hat{\psi}_k^{(-v)}$ as a fixed function of data from the complement of fold v.[5]

[5]For $k = \text{stack}$, $\hat{\psi}_k^{(-v)}$ denotes the fold-v stacked predictor $\hat{\alpha}^{(-v)\top}\hat{\psi}_{\text{tjs}}^{\text{oof}}(\cdot)$ from (A.19), which is a measurable function of the complement of fold v since both the weights $\hat{\alpha}^{(-v)}$ and the OOF base learners entering $\hat{\psi}_{\text{tjs}}^{\text{oof}}(X_i) = \big(\hat{\psi}_{\text{tf}}^{\text{oof}}(X_i),\, \hat{\psi}_{\text{joint}}^{\text{oof}}(X_i),\, \hat{\psi}_{\text{sep}}^{\text{oof}}(X_i)\big)^\top$ are trained on strict subsets thereof.

Then the discrete super learner $\hat{k}_n$ defined in (11) satisfies

$$\mathbb{E}\Big[\bar{R}(\hat{\psi}_{\hat{k}_n})\Big] \;\leq\; \min_{k\in\{1,\ldots,K\}} \mathbb{E}\Big[\bar{R}(\hat{\psi}_k)\Big] \;+\; \mathrm{O}\big(n^{-1/2}\big)\,. \tag{13}$$

The proof is given in Appendix A.3.

The $\mathrm{O}(n^{-1/2})$ remainder reflects sampling variation across the finite library of $K = 4$ candidates and does not involve nuisance estimation error, which is of strictly smaller order $\mathrm{o}(n^{-1/2})$ by the Neyman orthogonality of the R-loss. Because K is fixed, the remainder involves only a $\log K$ factor and requires no complexity regularization, entropy integral, or localized Rademacher complexity argument that would arise with a growing library (M. J. van der Laan & Dudoit, 2003; van der Vaart & Wellner, 1996).

6 Simulation

We evaluate the finite-sample performance of the three candidate learners and the super learner in a controlled simulation study. The main objective is to show that the super learner reliably identifies the best-performing candidate across settings in which each individual strategy is designed to dominate, and that it does not meaningfully sacrifice precision relative to the oracle-best learner in any scenario.

Implementation. While the super learner framework is agnostic to the choice of base learner, we implement all three strategies using Generalized Random Forests (GRF) (Tibshirani, Athey, & Wager, 2020), which minimize the R-loss by default and are therefore a natural fit for our framework. GRF derives Neyman-orthogonal forest weights by identifying the closest neighbors within the same leaf and combines them in a residual-on-residual regression following Robinson's partially linear model (Nie & Wager, 2020; Robinson, 1988). The forests are honest in the sense that the subsamples used for growing the trees and for estimating the leaf weights do not overlap, which guards against overfitting. The discrete super learner selects the candidate with the lowest cross-validated empirical risk as defined in (11). We additionally include the continuously stacked ensemble as a candidate in the library, since it can exploit convex combinations of the three strategies that the discrete selector cannot. The stacked weights are estimated by constrained least squares as described in Section 5.

Design. We consider four scenarios that differ in the complexity of the treatment effect functions and the nuisance structure; Table 2 provides the exact specifications. All covariates are drawn independently from Unif$(0, 1)$, with $p_\tau = 2$ treatment effect covariates, $p_{\text{nuis}} = 2$ nuisance covariates, and $p_{\text{irr}} = 5$ irrelevant noise covariates appended to the design matrix. We set $n \in \{500, 2{,}000\}$ and draw $R = 100$ independent replications per configuration. Performance is evaluated on a held-out test sample of $n_{\text{test}} = 1{,}000$ observations drawn from the same distribution. We report the empirical risk and the mean squared error (MSE) of each learner's predictions against the true $\psi^*(x)$, averaged across replications, as well as the selection frequency of the discrete super learner.

Scenarios. The four scenarios are constructed such that each individual strategy has at least one setting in which it is theoretically favoured, while the super learner is designed to perform well across all of them.

Scenario 1 features an aligned nuisance structure in which $\mu_Z(x) = \mu_Y(x) - \mu_C(x)$ is linear, so estimating the net nuisance directly is easy. The net effect $\psi(x)$ is simpler than either component effect individually, as τ_Y and τ_C share the same nonlinear basis and nearly cancel. This favours Strategy I (Transformed), which collapses the problem into a single composite outcome and directly exploits the simplicity of ψ.

Scenario 2 places the outcome and cost effects on orthogonal covariate subspaces: τ_Y depends only on X_1 and τ_C only on X_2. Any estimator that pools the two effects — as Strategies I and II do — cannot separately adapt to each subspace. Strategy III (Separate) is the only approach that can exploit this structure by fitting independent forests for each component, making it the natural winner here.

Scenario 3 combines a divergent nuisance structure, where μ_Y and μ_C depend on different nonlinear bases making μ_Z complex, with positively correlated errors ($\rho = 0.8$). The correlated noise reduces the effective variance of the joint residual relative to the sum of two independent residuals, which benefits Strategy II (Joint). By retaining separate nuisance models for Y and C while still targeting ψ in a single loss, Strategy II avoids the misspecification of Strategy I's composite nuisance while not accumulating the independent error variance of Strategy III.

Scenario 4 features a homogeneous net effect ($\psi(x) = 0.3$ for all x) despite complex and individually varying τ_Y and τ_C. No single inductive bias is unambiguously favoured: Strategy I benefits from the constant target but struggles with the complex nuisance; Strategy III is penalised by accumulating error from two separate nonlinear problems. This intermediate setting is precisely where the stacked ensemble can add value beyond discrete winner selection.

Results. Table 3 reports the results for $n = 2{,}000$; the $n = 500$ results confirm the same qualitative patterns at larger variance, consistent with $\sqrt{n}$-convergence of the nuisance estimators.

In Scenario 1, the Transformed and Joint learners achieve nearly identical empirical risk and MSE, both substantially outperforming Separate (MSE 0.013 and 0.014 versus 0.044). The super learner correctly concentrates selection on Joint and Transformed, picking Joint in 62% of replications. In Scenario 2, Separate wins decisively (MSE 0.032 versus 0.052–0.053 for the remaining learners), and the super learner selects it in 92% of replications, confirming that the empirical risk criterion reliably identifies the orthogonal structure. In Scenario 3, Joint achieves the lowest MSE (0.010) and is selected in 77% of replications at $n = 2{,}000$, with selection concentrating as the sample size grows and nuisance estimation improves.

Scenario 4 provides the clearest illustration of the ensemble's added value. The SL stack achieves an MSE of 0.003 at $n = 2{,}000$, compared to 0.006 for Transformed and 0.014 for Joint, and is selected by the discrete super learner in 62% of replications. This is the only scenario in which the stacked combination materially outperforms all individual candidates rather than merely recovering the best one, reflecting the intermediate complexity of the target that no single strategy can fully exploit.

Across all scenarios, the SL winner matches or closely approaches the oracle-best individual learner, never being dominated by a substantial margin. The selection frequencies confirm that the empirical risk criterion succeeds in identifying the favoured strategy in each setting, providing empirical support for the theoretical motivation of the ensemble.

Table 2: *Data-generating process specifications*

	Treatment effects			*Nuisance*		
Sc.	$\tau_Y(x)$	$\tau_C(x)$	$\psi(x)$	$\mu_Y(x)$	$\mu_C(x)$	μ_Z
1	$\phi(x)$	$\phi(x) - \frac{1}{2}(x_1 + x_2) + 0.3$	$\frac{1}{2}(x_1 + x_2) - 0.3$	$\sum_k g_k(x_k) + \ell(x)$	$\sum_k g_k(x_k)$	linear
2	$1.5 + 2 \cdot \mathbf{1}_{x_1>0.5}$	$3 \cdot \mathbf{1}_{x_2>0.5}$	$\tau_Y - \tau_C$	$\sum_k f_k(x_k)$	$\sum_k h_k(x_k)$	nonlinear
3	$1 + x_1 + x_2$	$1.3 + \frac{1}{2}(x_1 + x_2)$	$\frac{1}{2}(x_1 + x_2) - 0.3$	$\sum_k f_k(x_k)$	$\sum_k h_k(x_k)$	nonlinear
4	$2\sin 4x_1 + 1.5\cos 3x_2 + x_1x_2$	$\tau_Y - 0.3$	0.3	$\sum_k f_k(x_k)$	$\sum_k h_k(x_k)$	nonlinear

Notes: Sc. 1: aligned nuisance, μ_Z linear (favours Strategy I). Sc. 2: orthogonal τ_Y, τ_C on disjoint covariates (favours Strategy III). Sc. 3: divergent nuisance, correlated errors $\rho = 0.8$ (favours Strategy II). Sc. 4: homogeneous ψ, complex components (favours ensemble). ($p_\tau = 2$, $p_{\text{nuis}} = 2$). All covariates $X_k \sim \text{Unif}(0, 1)$. $h_1(x) = 3x^3 - 2x$, $h_2(x) = \cos(2\pi x)$.

Table 3: *Simulation results*

Sim.-par.		*Emp. Risk*					*MSE*					*Sel. Freq.*			
	N	I. tf	II. joint	III. sep	SL stack	SL winner	I. tf	II. joint	III. sep	SL stack	SL winner	I. tf	II. joint	III. sep	SL stack
Sc. 1	500	0.514	**0.513**	0.523	0.514	0.514	0.029	**0.027**	0.064	0.030	0.030	22%	62%	2%	14%
	2,000	0.506	**0.506**	0.513	0.506	0.507	0.013	0.014	0.044	**0.012**	0.014	37%	62%	0%	1%
Sc. 2	500	1.575	1.578	**1.530**	1.532	1.531	0.316	0.324	**0.126**	0.132	0.131	0%	0%	90%	10%
	2,000	1.007	1.007	**1.002**	1.003	1.002	0.053	0.052	**0.032**	0.034	0.033	6%	1%	92%	1%
Sc. 3	500	0.981	**0.980**	0.982	0.983	0.982	0.036	**0.033**	0.037	0.043	0.037	24%	30%	15%	31%
	2,000	0.482	**0.481**	0.481	0.481	0.481	0.014	**0.010**	0.012	0.011	0.011	5%	77%	14%	4%
Sc. 4	500	1.389	1.391	1.411	**1.388**	1.391	0.021	0.031	0.107	**0.017**	0.026	27%	31%	2%	40%
	2,000	0.885	0.886	0.894	**0.884**	0.884	0.006	0.014	0.050	**0.003**	0.005	29%	9%	0%	62%

Notes: ($p_{\text{nuis}} = 2$, $p_{\text{irr}} = 5$, $R = 100$ replications). Means across replications. Bold marks the lowest value among individual learners per metric and setting. Sel. Freq. is the share of replications in which the super learner selected each individual candidate.

7 Pledge donations

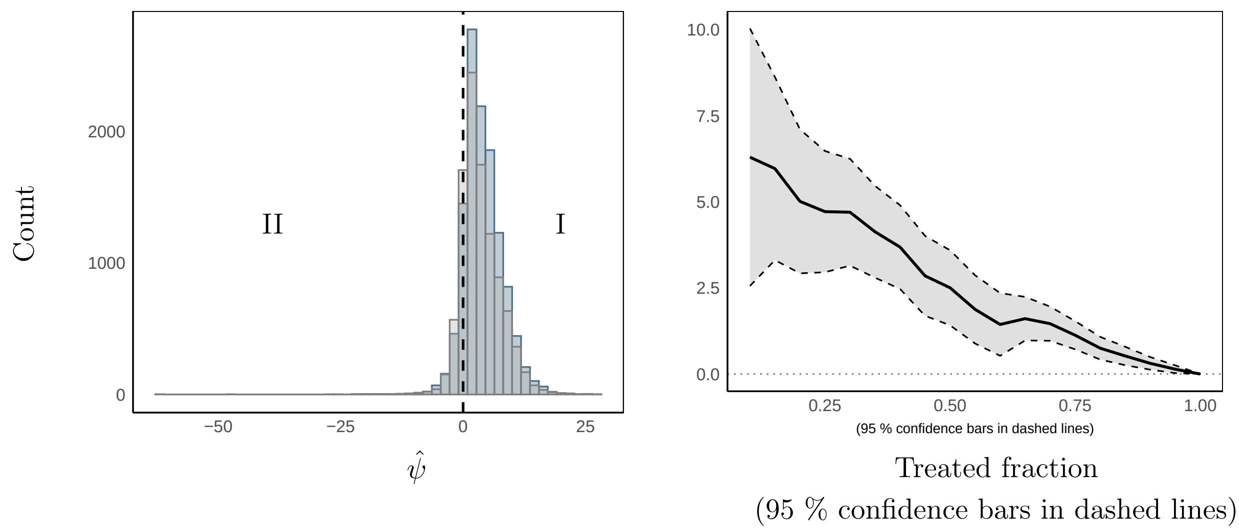
We apply our method to data from a fundraising campaign conducted by a non-profit organization (NGO). The NGO contacts pledge donors by letter, asking them to increase their recurring donations. This intervention could have two opposing effects: first, some donors may respond as intended and increase their payments; second, the mailing may serve as a reminder of their standing order and lead to cancellations. The objective is therefore to maximize donations while minimizing the risk of cancellations through targeted treatment assignment.

Data. The sample consists of$N = 22,038$ pledge donors who contribute on a quarterly or monthly basis. We control for donor characteristics and the past donor journey. The latter captures previous interactions, such as the date and amount of one-time donations or any earlier communication. All donors who receive the mailing have been signed up for at least two years. We use two donor cohorts: the treatment group received the mailing in March 2024, and the control group one month later. The outcome is the increase in pledge donations over the three months following assignment, and the cost variable is the amount of pledge cancellations in the same period. Both are measured in euros.

Estimation. We select the winner among the four candidate learners by minimizing the empirical risk in (11). The empirical risks are 902.07 (I. transformed), 902.32 (II. joint), 902.19 (III. separate), 902.10 (SL stack), selecting the transformed R-learner as the winner. The similarity of the empirical risks across candidates suggests that the net effect is not substantially more complex than its outcome and cost components individually, which favors direct targeting of $\psi^*(x)$.

Figure 2: *CNATE results for fundraising campaign*

(a) Histogram CNATE *(b) Targeting Operator Characteristics*



Notes: The figure summarizes the results for $\hat{\psi}(x)$. Sub-figure (a) shows the distribution of the CNATE, where W_i corresponds to receiving a letter for increasing the pledge donations, Y_i is the pledge upgrade, and C_i is the pledge cancellations, both measured in euros; sub-figure (b) shows the targeting operator characteristic with 95%-confidence bands based on influence-function standard errors.

We utilize the distribution of estimated net effects to evaluate the impact of the mailing. For inference, we trim observations with estimated propensity scores outside the interval $(0.025, 0.975)$ to ensure stable inverse probability weighting. We then assess overall targeting quality using the rank average treatment effect (Yadlowsky et al., 2025), which summarizes the gain from prioritizing units by their estimated CNATE relative to uniform treatment. The point estimate is 2.833 with a standard error of 0.564, confirming a statistically significant positive effect of the priority score.

Results. The results are graphically summarized in Figure 2. The target group that should receive the mailing is identified by the fraction of donors with a positive CNATE, group I in subfigure 2. This group includes donors for whom the outcome effect outweighs the cost effect. In contrast, the group of donors with a negative CNATE, group II, should not be targeted. The targeting operator characteristic in subfigure 2 depicts the gain from treating subgroups ranked by their estimated net effect, relative to the average effect from treating the entire sample. The curve exhibits an overall downward trend with minor local fluctuations and lies significantly above zero for treated fractions up to roughly 0.85, indicating that donors can be meaningfully ranked by their net benefit. For instance, treating only the top 20% of donors by estimated net effect yields a substantially higher average gain than treating the full sample, illustrating the welfare improvements achievable through targeted assignment.

8 Conclusion

We e propose a super learner for estimating heterogeneous net treatment effects under unit-varying outcome and cost effects of a binary treatment. The key challenge is that the functional complexity of the net effect and its underlying components is unknown in any given application, and the appropriate level of aggregation, whether to target the net effect directly or to estimate its components separately, depends on this unknown structure. We formalize this challenge as a problem of inductive bias and show that each of the three R-learner strategies we consider can be optimal under a specific data-generating process.

Our simulation study demonstrates this across four scenarios designed so that each individual strategy dominates the others in at least one setting. Directly targeting the net effect via the transformed R-learner performs best when the outcome and cost effects share similar functional form and largely cancel, yielding a simpler net effect. The separate R-learner is advantageous when the outcome and cost effects are driven by orthogonal covariates, so that the net effect is more complex than either component alone. The joint R-learner occupies an intermediate position, retaining separate nuisance models while targeting the net effect in a single loss. By combining all three strategies through constrained stacking and selecting the discrete winner by minimising cross-validated empirical risk, the ensemble is competitive across all scenarios and strictly dominates any single strategy on average.

In an empirical application using data from a fundraising campaign run by a large non-profit organisation, the transformed R-learner is selected as the winner, with empirical risks across all candidates differing only marginally. This finding suggests that the net effect of the mailing — the difference between pledge upgrades and cancellations — is not substantially more complex than its components individually, so that direct targeting is sufficient. The rank average treatment effect estimate of 2.833 (SE 0.564) confirms that the estimated net effects meaningfully discriminate between donors, and the targeting operator characteristic shows that concentrating the mailing on the highest-ranked donors yields substantially larger gains than uniform targeting.

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A Appendix

A.1 Identification of CNATE

By linearity of expectation, unconfoundedness (Assumption 1(a)), and SUTVA (Assumption 1(c)):

$$
\begin{aligned}
\psi^*(x) &= \mathbb{E}\big[\{Y_i(1) - Y_i(0)\} - \{C_i(1) - C_i(0)\} \;\big|\; X_i = x\big] \\
&= \mathbb{E}[Y_i(1) \mid X_i = x] - \mathbb{E}[Y_i(0) \mid X_i = x] - \mathbb{E}[C_i(1) \mid X_i = x] + \mathbb{E}[C_i(0) \mid X_i = x] \\
&= \mathbb{E}[Y_i(1) \mid W_i = 1,\, X_i = x] - \mathbb{E}[Y_i(0) \mid W_i = 0,\, X_i = x] \\
&\quad - \mathbb{E}[C_i(1) \mid W_i = 1,\, X_i = x] + \mathbb{E}[C_i(0) \mid W_i = 0,\, X_i = x] \\
&= \mathbb{E}[Y_i \mid W_i = 1,\, X_i = x] - \mathbb{E}[Y_i \mid W_i = 0,\, X_i = x] \\
&\quad - \mathbb{E}[C_i \mid W_i = 1,\, X_i = x] + \mathbb{E}[C_i \mid W_i = 0,\, X_i = x], \qquad \text{(A.1)}
\end{aligned}
$$

where each conditional expectation is well-defined by the overlap assumption (Assumption 1(b)), and the potential outcomes are individual-specific by no interference (Assumption 1(c)). The right-hand side of (A.1) depends only on the distribution of the observed data O_i, so $\psi^*(x)$ is identified. □

A.2 Neyman orthogonality for the score of CNATE

We show that the score associated with the conditional net average treatment effect (CNATE) satisfies the Neyman orthogonality condition (Chernozhukov et al., 2018), ensuring that the CNATE estimator attains its oracle nonparametric rate under slow (product-rate) nuisance convergence. Throughout this subsection, all residuals are population residuals at the true nuisances, marked with a star to distinguish them from the estimated OOF residuals used in Section 5.

Setup. Following the partially linear regression approach of Robinson (1988), we posit

$$Y_i = W_i\,\tau_Y^*(X_i) + g_Y^*(X_i) + \varepsilon_{Y,i}, \tag{A.2}$$

$$C_i = W_i\,\tau_C^*(X_i) + g_C^*(X_i) + \varepsilon_{C,i}, \tag{A.3}$$

with $\mathbb{E}[\varepsilon_{Y,i} \mid W_i, X_i] = \mathbb{E}[\varepsilon_{C,i} \mid W_i, X_i] = 0$. Under Assumption 1 (unconfoundedness, overlap, SUTVA), the CNATE is

$$\psi^*(X_i) \;=\; \tau_Y^*(X_i) - \tau_C^*(X_i).$$

In addition to the nuisances $\mu_Y^*(X_i) = \mathbb{E}[Y_i \mid X_i]$ and $\mu_C^*(X_i) = \mathbb{E}[C_i \mid X_i]$ already introduced in the main text, we use the propensity score $e^*(X_i) = \Pr(W_i = 1 \mid X_i)$, collected in $\nu^* = (\mu_Y^*, \mu_C^*, e^*)$. Define the population residuals

$$\tilde{Y}_i^* = Y_i - \mu_Y^*(X_i), \qquad \tilde{C}_i^* = C_i - \mu_C^*(X_i), \qquad \tilde{W}_i^* = W_i - e^*(X_i).$$

Solving (A.2)–(A.3) for the confounding functions g_Y^*, g_C^* and substituting back yields the Robinson-style residualized equations

$$\tilde{Y}_i^* = \tilde{W}_i^*\,\tau_Y^*(X_i) + \varepsilon_{Y,i}, \tag{A.4}$$

$$\tilde{C}_i^* = \tilde{W}_i^*\,\tau_C^*(X_i) + \varepsilon_{C,i}. \tag{A.5}$$

Subtracting, multiplying by $\tilde{W}_i^*$, and taking expectations conditional on X_i gives the moment condition underlying the R-learner of Nie and Wager (2020):

$$\mathbb{E}\Big[\tilde{W}_i^* \Big\{(\tilde{Y}_i^* - \tilde{C}_i^*) - \psi^*(X_i)\, \tilde{W}_i^*\Big\}\Big] = 0. \tag{A.6}$$

The corresponding score admits the linear-in-ψ^* decomposition

$$\varphi_{\psi^*,\,\nu^*}(O_i) \;=\; \underbrace{\big(-\tilde{W}_i^{*\,2}\big)}_{\varphi^A_{\nu^*}(O_i)}\, \psi^*(X_i) \;+\; \underbrace{\tilde{W}_i^*\, (\tilde{Y}_i^* - \tilde{C}_i^*)}_{\varphi^B_{\nu^*}(O_i)}. \tag{A.7}$$

(a) Unbiasedness of the score. Using $\mathbb{E}[\varepsilon_{Y,i}\tilde{W}_i^* \mid X_i] = \mathbb{E}[\varepsilon_{C,i}\tilde{W}_i^* \mid X_i] = 0$ and the law of iterated expectations,

$$\mathbb{E}\Big[\tilde{W}_i^*(\tilde{Y}_i^* - \tilde{C}_i^*)\Big] \;=\; \mathbb{E}\Big[\psi^*(X_i)\, \tilde{W}_i^{*\,2}\Big],$$

so that $\mathbb{E}[\varphi_{\psi^*,\,\nu^*}(O_i)] = 0$, confirming (A.6).

(b) Linearity in ψ^*. By construction of (A.7), the score has the linear form $\varphi_{\psi^*,\,\nu^*}(O_i) = \varphi^A_{\nu^*}(O_i)\, \psi^*(X_i) + \varphi^B_{\nu^*}(O_i)$, where neither $\varphi^A_{\nu^*}$ nor $\varphi^B_{\nu^*}$ depends on ψ^*. This is the "linear scores" structure of Chernozhukov et al. (2018).

(c) Neyman orthogonality. Consider Gateaux perturbations along arbitrary directions $(h_{\mu_Y}, h_{\mu_C}, h_e)$:

$$\mu_{Y,r} = \mu_Y^* + r\, h_{\mu_Y}, \qquad \mu_{C,r} = \mu_C^* + r\, h_{\mu_C}, \qquad e_r = e^* + r\, h_e,$$

and write $\nu_r = (\mu_{Y,r}, \mu_{C,r}, e_r)$, with corresponding perturbed residuals

$$\tilde{Y}_{i,r} = Y_i - \mu_{Y,r}(X_i), \qquad \tilde{C}_{i,r} = C_i - \mu_{C,r}(X_i), \qquad \tilde{W}_{i,r} = W_i - e_r(X_i).$$

Note that $\tilde{Y}_{i,0} = \tilde{Y}_i^*$, $\tilde{C}_{i,0} = \tilde{C}_i^*$, and $\tilde{W}_{i,0} = \tilde{W}_i^*$, and that

$$\frac{d}{dr}\tilde{Y}_{i,r}\bigg|_{r=0} = -h_{\mu_Y}(X_i), \quad \frac{d}{dr}\tilde{C}_{i,r}\bigg|_{r=0} = -h_{\mu_C}(X_i), \quad \frac{d}{dr}\tilde{W}_{i,r}\bigg|_{r=0} = -h_e(X_i). \tag{A.8}$$

Substituting the perturbed nuisances into (A.7),

$$\varphi_{\psi^*,\nu_r}(O_i) \;=\; \tilde{W}_{i,r}\big(\tilde{Y}_{i,r} - \tilde{C}_{i,r}\big) \;-\; \psi^*(X_i)\,\tilde{W}_{i,r}^2. \tag{A.9}$$

Differentiating (A.9) with respect to r via the product and chain rules,

$$\begin{aligned}
\frac{d}{dr}\,\varphi_{\psi^*,\nu_r}(O_i) &= \frac{d}{dr}\Big[\tilde{W}_{i,r}\big(\tilde{Y}_{i,r} - \tilde{C}_{i,r}\big)\Big] \;-\; \psi^*(X_i)\,\frac{d}{dr}\big[\tilde{W}_{i,r}^2\big] \\
&= \left(\frac{d\tilde{W}_{i,r}}{dr}\right)\big(\tilde{Y}_{i,r} - \tilde{C}_{i,r}\big) + \tilde{W}_{i,r}\left(\frac{d\tilde{Y}_{i,r}}{dr} - \frac{d\tilde{C}_{i,r}}{dr}\right) \\
&\quad - 2\,\psi^*(X_i)\,\tilde{W}_{i,r}\,\frac{d\tilde{W}_{i,r}}{dr}.
\end{aligned} \tag{A.10}$$

Evaluating at $r = 0$ and inserting the directional derivatives from (A.8),

$$\begin{aligned}\frac{d}{dr}\,\varphi_{\psi^*,\nu_r}(O_i)\Big|_{r=0} &= \big(-h_e(X_i)\big)\big(\tilde{Y}_i^* - \tilde{C}_i^*\big) + \tilde{W}_i^*\big(-h_{\mu_Y}(X_i) - (-h_{\mu_C}(X_i))\big) \\ &\quad - 2\,\psi^*(X_i)\,\tilde{W}_i^*\big(-h_e(X_i)\big) \\ &= -h_e(X_i)\big(\tilde{Y}_i^* - \tilde{C}_i^*\big) + \tilde{W}_i^*\big(-h_{\mu_Y}(X_i) + h_{\mu_C}(X_i)\big) \\ &\quad + 2\,\psi^*(X_i)\,\tilde{W}_i^*\,h_e(X_i). \end{aligned} \tag{A.11}$$

Taking expectations of (A.11) and conditioning on X_i via the law of iterated expectations, the first term factors out $h_e(X_i)$ (a function of X_i alone) and the second and third terms factor out $\tilde{W}_i^*$:

$$\begin{aligned}\frac{d}{dr}\,\mathbb{E}[\varphi_{\psi^*,\nu_r}(O_i)]\Big|_{r=0} &= \mathbb{E}\Big[-h_e(X_i)\big(\tilde{Y}_i^* - \tilde{C}_i^*\big)\Big] \\ &\quad + \mathbb{E}\Big[\tilde{W}_i^*\big(-h_{\mu_Y}(X_i) + h_{\mu_C}(X_i)\big)\Big] + \mathbb{E}\Big[2\,\psi^*(X_i)\,\tilde{W}_i^*\,h_e(X_i)\Big] \\ &= \underbrace{\mathbb{E}\Big[-h_e(X_i)\,\mathbb{E}\big[\tilde{Y}_i^* - \tilde{C}_i^* \,\big|\, X_i\big]\Big]}_{T_1} \\ &\quad + \underbrace{\mathbb{E}\Big[\big(-h_{\mu_Y}(X_i) + h_{\mu_C}(X_i) + 2\psi^*(X_i)h_e(X_i)\big)\,\mathbb{E}\big[\tilde{W}_i^* \,\big|\, X_i\big]\Big]}_{T_2}. \end{aligned} \tag{A.12}$$

By construction $\mathbb{E}[\tilde{Y}_i^* \mid X_i] = \mathbb{E}[\tilde{C}_i^* \mid X_i] = 0$ and $\mathbb{E}[\tilde{W}_i^* \mid X_i] = 0$, so the inner conditional expectations in both T_1 and T_2 vanish, giving $T_1 = T_2 = 0$. The map $\nu \mapsto \mathbb{E}[\varphi_{\psi^*,\nu}(O_i)]$ is twice continuously Gateaux-differentiable, and the Neyman orthogonality condition holds (Chernozhukov et al., 2018).

(d) Regularity and convergence rate. Assume the propensity score is bounded, $e^*(X_i) \in [\varepsilon, 1-\varepsilon]$ for some $\varepsilon > 0$ (overlap; Assumption 1 (b)), and that (Y_i, C_i, X_i) have sub-Gaussian tails, so the score in (A.7) is measurable and square-integrable. Provided the nuisance estimators $\hat{\nu} = (\hat{\mu}_Y, \hat{\mu}_C, \hat{e})$ satisfy the product-rate conditions

$$\|\hat{\mu}_Y - \mu_Y^*\|_2 \cdot \|\hat{e} - e^*\|_2 = \mathrm{o}_p(n^{-1/2}), \qquad \|\hat{\mu}_C - \mu_C^*\|_2 \cdot \|\hat{e} - e^*\|_2 = \mathrm{o}_p(n^{-1/2}), \tag{A.13}$$

(typically met when each nuisance converges faster than $n^{-1/4}$), Neyman orthogonality combined with cross-fitting ensures that the first-order bias from nuisance estimation vanishes and that the remaining second-order bias is negligible in large samples. Unlike a finite-dimensional target, however, the CNATE $\psi^*(\cdot)$ is itself an unknown function of covariates, so the second-stage R-learner converges at a nonparametric rate that depends on the smoothness or structural complexity of ψ^* and on the second-stage estimator employed. What orthogonality delivers is the oracle property: the excess risk of the estimated CNATE matches, up to lower-order terms, the rate that would be achieved if the true nuisances were known (Foster & Syrgkanis, 2023; Kennedy, 2023; Nie & Wager, 2020).

A.3 Proof of Theorem 1

Notation. Let $O_i = (X_i, Y_i, C_i, W_i)$ and $Z_i = Y_i - C_i$. Write $\nu^* = (\mu_Z^*, e^*)$ for the true nuisances and, for each fold v, $\hat{\nu}^{(-v)} = (\hat{\mu}_Z^{(-v)}, \hat{e}^{(-v)})$ for their cross-fitted estimates on the training complement $\mathcal{D}^{(-v)} := \{O_i : i \notin \mathrm{Val}(v)\}$, with folds $v = 1, \ldots, V$ indexing validation sets $\mathrm{Val}(v)$ and $v(i)$ denoting the fold containing observation i.

Following Section 5, residuals come in two flavors. The estimated OOF residuals, $\tilde{Z}_i = Z_i - \hat{\mu}_Z^{(-v(i))}(X_i)$ and $\tilde{W}_i = W_i - \hat{e}^{(-v(i))}(X_i)$ are those defined in (6) and used to score candidates in (10). The population residuals, $\tilde{Z}_i^* = Z_i - \mu_Z^*(X_i)$ and $\tilde{W}_i^* = W_i - e^*(X_i)$, are their counterparts at the true nuisances and enter the population R-risk. For any ψ and $\nu = (\mu_Z, e)$, define the R-loss

$$\ell_\nu(O_i, \psi) \;=\; \big((Z_i - \mu_Z(X_i)) - (W_i - e(X_i))\psi(X_i)\big)^2,$$

so that $\ell_{\hat{\nu}^{(-v(i))}}(O_i, \psi) = (\tilde{Z}_i - \tilde{W}_i\,\psi(X_i))^2$ and $\ell_{\nu^*}(O_i, \psi) = (\tilde{Z}_i^* - \tilde{W}_i^*\,\psi(X_i))^2$. For a fixed function ψ, write $R(\psi) = \mathbb{E}[\ell_{\nu^*}(O_i, \psi)]$ for the population R-risk at true nuisances, and define the fold-averaged population risk of a cross-fitted estimator by

$$\bar{R}(\hat{\psi}_k) := \frac{1}{V}\sum_{v=1}^{V} R(\hat{\psi}_k^{(-v)}).$$

The cross-validated empirical risk, reproducing (10), is

$$\hat{R}_{\text{CV}}(\hat{\psi}_k) \;=\; \frac{1}{V}\sum_{v=1}^{V}\frac{1}{n_v}\sum_{i\in\text{Val}(v)} \ell_{\hat{\nu}^{(-v)}}(O_i, \hat{\psi}_k^{(-v)}) \;=\; \frac{1}{n}\sum_{i=1}^{n}\big(\tilde{Z}_i - \tilde{W}_i\,\hat{\psi}_k^{\text{oof}}(X_i)\big)^2,$$

where the second equality uses $v = v(i)$ for $i \in \text{Val}(v)$ and $\hat{\psi}_k^{\text{oof}}$ from (8). We write $\hat{R}_{\text{CV}}(\hat{\psi}_k; \nu^*)$ for the same quantity with $\hat{\nu}^{(-v)}$ replaced by ν^*, i.e.,

$$\hat{R}_{\text{CV}}(\hat{\psi}_k; \nu^*) = \frac{1}{V}\sum_{v=1}^{V}\frac{1}{n_v}\sum_{i\in\text{Val}(v)} \ell_{\nu^*}(O_i, \hat{\psi}_k^{(-v)}).$$

The cross-validated risk $\hat{R}_{\text{CV}}$ uses the transformed-loss residualization $(\hat{\mu}_Z^{(-v)}, \hat{e}^{(-v)})$ for all four candidates, regardless of which R-learner loss each candidate was trained on. The proof therefore analyzes a single population functional R and a single empirical functional $\hat{R}_{\text{CV}}$, both built from the transformed loss, and shows that $\hat{\psi}_{\text{tf}}$, $\hat{\psi}_{\text{joint}}$, $\hat{\psi}_{\text{sep}}$, and $\hat{\psi}_{\text{stack}}$

(viewed, conditional on $\mathcal{D}^{(-v)}$, as fixed functions) all compete against this common criterion. That $\hat{\psi}_{\text{joint}}$ and $\hat{\psi}_{\text{sep}}$ are trained using different losses (R_{joint} and the pair (R_Y, R_C) from Table 1) does not affect the scoring step: the transformed population functional coincides at the truth with the objectives used for training, since R_{joint} targets the same population object ($\mu_Z^* = \mu_Y^* - \mu_C^*$) and (R_Y, R_C) target it via $\psi^* = \tau_Y^* - \tau_C^*$.

The proof follows the general procedure in L. van der Laan (2026). First, we establish the basic inequality $\bar{R}(\hat{\psi}_{\hat{k}_n}) - \bar{R}(\hat{\psi}_{k^*}) \le 2\max_k |D_k|$ from the optimality of $\hat{k}_n$ in (11), where $k^* = \arg\min_k \mathbb{E}[\bar{R}(\hat{\psi}_k)]$ is the deterministic oracle index. We then bound $\max_k |D_k|$ by splitting it into a nuisance term $\Delta_k = \mathrm{o}_p(n^{-1/2})$ and a statistical fluctuation term $\Gamma_k = \mathrm{O}_p(n^{-1/2})$, which assemble into the oracle inequality in (13).

Basic inequality. Since $\hat{k}_n$ minimizes $\hat{R}_{\text{CV}}$ by (11), $\hat{R}_{\text{CV}}(\hat{\psi}_{\hat{k}_n}) \le \hat{R}_{\text{CV}}(\hat{\psi}_{k^*})$. Adding and subtracting $\bar{R}(\hat{\psi}_{\hat{k}_n})$ and $\bar{R}(\hat{\psi}_{k^*})$ and rearranging gives

$$\bar{R}(\hat{\psi}_{\hat{k}_n}) - \bar{R}(\hat{\psi}_{k^*}) \;\le\; \underbrace{\left[\hat{R}_{\text{CV}}(\hat{\psi}_{k^*}) - \bar{R}(\hat{\psi}_{k^*})\right]}_{D_{k^*}} - \underbrace{\left[\hat{R}_{\text{CV}}(\hat{\psi}_{\hat{k}_n}) - \bar{R}(\hat{\psi}_{\hat{k}_n})\right]}_{D_{\hat{k}_n}} \;\le\; 2\max_k |D_k|, \tag{A.14}$$

so it suffices to show $\max_k |D_k| = \mathrm{O}_p(n^{-1/2})$. Decompose

$$D_k \;=\; \underbrace{\left[\hat{R}_{\text{CV}}(\hat{\psi}_k) - \hat{R}_{\text{CV}}(\hat{\psi}_k; \nu^*)\right]}_{\Delta_k} \;+\; \underbrace{\left[\hat{R}_{\text{CV}}(\hat{\psi}_k; \nu^*) - \bar{R}(\hat{\psi}_k)\right]}_{\Gamma_k}. \tag{A.15}$$

We bound Δ_k and Γ_k separately.

Nuisance approximation error is second-order. We show $|\Delta_k| = o_p(n^{-1/2})$ for each k. From (A.15),

$$\Delta_k \;=\; \frac{1}{V}\sum_{v=1}^{V}\frac{1}{n_v}\sum_{i\in \mathrm{Val}(v)}\big[\ell_{\hat{\nu}^{(-v)}}(O_i,\hat{\psi}_k^{(-v)}) - \ell_{\nu^*}(O_i,\hat{\psi}_k^{(-v)})\big], \tag{A.16}$$

the empirical average of the per-observation deviation between the loss at estimated and at true nuisances. Fix a fold v. Conditional on $\mathcal{D}^{(-v)}$, both $\hat{\nu}^{(-v)}$ and $\hat{\psi}_k^{(-v)}$ are fixed, so a second-order Taylor expansion of $\ell_\nu(O_i,\hat{\psi}_k^{(-v)})$ in ν around ν^* gives

$$\ell_{\hat{\nu}^{(-v)}}(O_i,\hat{\psi}_k^{(-v)}) - \ell_{\nu^*}(O_i,\hat{\psi}_k^{(-v)}) \;=\; \mathrm{Lin}_i + \mathrm{Quad}_i,$$

where Lin_i is linear in the nuisance perturbation $h := (h_\mu, h_e) = (\hat{\mu}_Z^{(-v)} - \mu_Z^*,\, \hat{e}^{(-v)} - e^*)$ and Quad_i is the quadratic remainder. Direct computation gives

$$\mathrm{Lin}_i = -2\big(\tilde{Z}_i^* - \tilde{W}_i^*\,\hat{\psi}_k^{(-v)}(X_i)\big)\big(h_\mu(X_i) - \hat{\psi}_k^{(-v)}(X_i)\,h_e(X_i)\big).$$

The empirical average of Lin_i over $\mathrm{Val}(v)$ vanishes in expectation conditional on $\mathcal{D}^{(-v)}$ by Neyman orthogonality of the R-loss (Appendix A.2), which gives the per-observation identity $\mathbb{E}[\mathrm{Lin}_i \mid X_i, \mathcal{D}^{(-v)}] = 0$ via $\mathbb{E}[\tilde{Z}_i^* \mid X_i] = 0$ and $\mathbb{E}[\tilde{W}_i^* \mid X_i] = 0$. This is the first-order insensitivity condition of Chernozhukov et al. (2018). Hence $\mathbb{E}\Big[n_v^{-1}\sum_{i\in\mathrm{Val}(v)}\mathrm{Lin}_i \,\Big|\, \mathcal{D}^{(-v)}\Big] = 0$ by linearity of expectation.

The quadratic remainder satisfies, after averaging over $\mathrm{Val}(v)$ and applying Cauchy–

Schwarz,

$$\left|\frac{1}{n_v}\sum_{i\in\mathrm{Val}(v)}\mathrm{Quad}_i\right| \;\lesssim\; \|\hat{\mu}_Z^{(-v)}-\mu_Z^*\|_{L^2}^2+\|\hat{e}^{(-v)}-e^*\|_{L^2}^2+\|\hat{\mu}_Z^{(-v)}-\mu_Z^*\|_{L^2}\,\|\hat{e}^{(-v)}-e^*\|_{L^2}, \quad \text{(A.17)}$$

where the cross-product term arises from the product structure of the R-loss. Under Assumption 2(c), each squared term is $\mathrm{o}_p(n^{-1/2})$ and the cross-product term is $\mathrm{o}_p(n^{-1/2})$ by Cauchy–Schwarz, so the per-fold quadratic contribution is $\mathrm{o}_p(n^{-1/2})$. Averaging over the V (constant) folds preserves this rate, giving $|\Delta_k| = \mathrm{o}_p(n^{-1/2})$ for all k in (A.16).

Statistical fluctuation is $\mathrm{O}_p(n^{-1/2})$. It remains to bound Γ_k in (A.15), the centered deviation of the empirical risk at true nuisances from its fold-averaged population counterpart. For each fold v, conditional on $\mathcal{D}^{(-v)}$, the terms $\ell_{\nu^*}(O_i,\hat{\psi}_k^{(-v)}) = (\tilde{Z}_i^* - \tilde{W}_i^*\,\hat{\psi}_k^{(-v)}(X_i))^2$ for $i \in \mathrm{Val}(v)$ are i.i.d. and sub-exponential under Assumptions 2(a)–(b). A Bernstein inequality (L. van der Laan, 2026, Lemma 2) gives, for any $t > 0$,

$$\Pr\big(|\Gamma_k^{(v)}| > t \,\big|\, \mathcal{D}^{(-v)}\big) \;\le\; 2\exp\!\left(-\frac{n_v t^2}{2(\sigma_k^2 + Mt/3)}\right),$$

where $\sigma_k^2 = \mathrm{Var}(\ell_{\nu^*}(O_i,\hat{\psi}_k^{(-v)}) \mid \mathcal{D}^{(-v)})$ is bounded uniformly in v under Assumptions 2(a)–(b). A union bound over the V (constant) folds yields

$$\Pr(|\Gamma_k| > t) \;\le\; 2V\exp\!\left(-\frac{nt^2}{C_k}\right) \quad \text{(A.18)}$$

for a constant $C_k > 0$. Since $K = 4$ is fixed, a union bound over candidates gives

$$\Pr\!\left(\max_{k=1,\dots,K}|\Gamma_k| > t\right) \;\le\; 2VK\exp\!\left(-\frac{nt^2}{C}\right), \qquad C = \max_k C_k.$$

Setting $t_n = \sqrt{C\log(2VK)/n}$ yields $\max_k |\Gamma_k| = \mathrm{O}_p(\sqrt{\log K/n}) = \mathrm{O}_p(n^{-1/2})$, since K and V are fixed.

Assembling the bound. Recall k^* is deterministic with $k^* = \arg\min_k \mathbb{E}[\bar{R}(\hat{\psi}_k)]$. Combining (A.14)–(A.18),

$$\bar{R}(\hat{\psi}_{\hat{k}_n}) - \bar{R}(\hat{\psi}_{k^*}) \;\le\; 2\max_k |\Delta_k| + 2\max_k |\Gamma_k| \;=\; \mathrm{o}_p(n^{-1/2}) + \mathrm{O}_p(n^{-1/2}) \;=\; \mathrm{O}_p(n^{-1/2}).$$

By uniform integrability (which follows from the bounded predictions in Assumption 2(a)), the rescaled stochastic bounds upgrade to bounds in expectation: $\mathbb{E}[n^{1/2}|\Delta_k|] \to 0$ and $\mathbb{E}[n^{1/2}|\Gamma_k|] = \mathrm{O}(1)$. Taking expectations yields

$$\mathbb{E}\Big[\bar{R}(\hat{\psi}_{\hat{k}_n})\Big] - \min_k \mathbb{E}\Big[\bar{R}(\hat{\psi}_k)\Big] \;\le\; \mathrm{o}(n^{-1/2}) + \mathrm{O}(n^{-1/2}) \;=\; \mathrm{O}(n^{-1/2}),$$

which is the bound in (13). □

A.4 Extension: the stacked candidate

We consider the continuous super learner that combines the three base learners through a non-negative weighted average as in (9). Let $\hat{\psi}^{\mathrm{oof}}_{\mathrm{tjs}}(X_i) = \big(\hat{\psi}^{\mathrm{oof}}_{\mathrm{tf}}(X_i),\, \hat{\psi}^{\mathrm{oof}}_{\mathrm{joint}}(X_i),\, \hat{\psi}^{\mathrm{oof}}_{\mathrm{sep}}(X_i)\big)^\top$ denote the vector of OOF base-learner predictions from (8). The fold-v weights solve a ridge-regularized constrained least-squares problem on the complement with

$$\hat{\alpha}^{(-v)} = \arg\min_{\alpha \ge 0} \; \frac{1}{n - n_v} \sum_{i \notin \mathrm{Val}(v)} \Big(\tilde{Z}_i - \tilde{W}_i\, \alpha^\top \hat{\psi}^{\mathrm{oof}}_{\mathrm{tjs}}(X_i)\Big)^2 + \lambda \|\alpha\|_2^2, \qquad \text{(A.19)}$$

with $\alpha = (\alpha_1, \alpha_2, \alpha_3)^\top$ and a small ridge penalty $\lambda > 0$ for numerical stability. Because $v(i) \neq v$ for every $i \notin \mathrm{Val}(v)$, the inputs $\tilde{Z}_i$, $\tilde{W}_i$, and $\hat{\psi}^{\mathrm{oof}}_{\mathrm{tjs}}(X_i)$ are all out-of-sample for observation i: the stacking step optimizes over genuinely held-out base-learner predictions and nuisance residuals. The fold-v stacked prediction at its held-out fold is $\hat{\psi}^{(-v)}_{\mathrm{stack}}(X_i) = \hat{\alpha}^{(-v)\top}\hat{\psi}^{\mathrm{oof}}_{\mathrm{tjs}}(X_i)$ for $i \in \mathrm{Val}(v)$, yielding $\hat{\psi}^{\mathrm{oof}}_{\mathrm{stack}}(X_i)$ for all $i = 1, \ldots, n$ after pooling.

The argument in Appendix A.3 covers the three base learners $\hat{\psi}_{\mathrm{tf}}$, $\hat{\psi}_{\mathrm{joint}}$, $\hat{\psi}_{\mathrm{sep}}$ directly, since each $\hat{\psi}^{(-v)}_k$ is a fixed function of $\mathcal{D}^{(-v)}$. We now show that the bound extends to the stacked candidate. Recall from (A.19) that

$$\hat{\psi}^{(-v)}_{\mathrm{stack}}(x) \;=\; \hat{\alpha}^{(-v)\top}\hat{\psi}^{\mathrm{oof}}_{\mathrm{tjs}}(x),$$

where $\hat{\alpha}^{(-v)}$ solves the constrained least-squares problem (A.19) on $\mathcal{D}^{(-v)}$ and $\hat{\psi}^{\mathrm{oof}}_{\mathrm{tjs}} = (\hat{\psi}^{\mathrm{oof}}_{\mathrm{tf}}, \hat{\psi}^{\mathrm{oof}}_{\mathrm{joint}}, \hat{\psi}^{\mathrm{oof}}_{\mathrm{sep}})^\top$ collects the OOF base-learner predictions from (8). Each component $\hat{\psi}^{(-v(i))}_k$ entering the pool for $i \notin \mathrm{Val}(v)$ is trained on a strict subset of $\mathcal{D}^{(-v)}$, and $\hat{\alpha}^{(-v)}$ is itself a measurable function of $\mathcal{D}^{(-v)}$. Consequently $\hat{\psi}^{(-v)}_{\mathrm{stack}}$, restricted to $\mathrm{Val}(v)$, is a fixed function of $\mathcal{D}^{(-v)}$, and the conditional independence argument used for the base learners applies verbatim. The decomposition $D_{\mathrm{stack}} = \Delta_{\mathrm{stack}} + \Gamma_{\mathrm{stack}}$ from (A.15) still holds with both terms bounded as before. The only additional contribution comes from estimating $\hat{\alpha}^{(-v)}$ rather than using the population-optimal weights α^*, which we now bound.

Write

$$\hat{\psi}^{(-v)}_{\mathrm{stack}} \;=\; \alpha^{*\top}\hat{\psi}^{\mathrm{oof}}_{\mathrm{tjs}} + (\hat{\alpha}^{(-v)} - \alpha^*)^\top\hat{\psi}^{\mathrm{oof}}_{\mathrm{tjs}}.$$

The first summand is a fixed linear combination of base-learner predictions and is covered by the base-learner analysis. For the second, α^* is the population counterpart of $\hat{\alpha}^{(-v)}$, i.e., the non-negative minimiser of the population R-risk of the convex combination,

$$\alpha^* \;:=\; \arg\min_{\alpha \geq 0} \mathbb{E}\Big[\big(\tilde{Z}_i^* - \tilde{W}_i^*\, \alpha^\top \hat{\psi}_{\mathrm{tjs}}^{\mathrm{oof}}(X_i)\big)^2\Big].$$

Because α is constrained to the non-negative orthant, α^* satisfies the Karush–Kuhn–Tucker conditions rather than the unconstrained first-order conditions: $\nabla R(\alpha^*)_j = 0$ for j with $\alpha_j^* > 0$, $\nabla R(\alpha^*)_j \geq 0$ for j with $\alpha_j^* = 0$, together with complementary slackness $\nabla R(\alpha^*)_j\, \alpha_j^* = 0$. A second-order expansion of R around α^* yields

$$R(\hat{\psi}_{\mathrm{stack}}^{(-v)}) - R(\alpha^{*\top} \hat{\psi}_{\mathrm{tjs}}^{\mathrm{oof}}) \;=\; (\hat{\alpha}^{(-v)} - \alpha^*)^\top \nabla R(\alpha^*) + \mathrm{O}\big(\|\hat{\alpha}^{(-v)} - \alpha^*\|_2^2\big), \tag{A.20}$$

where, by complementary slackness, the linear term collapses to a sum over boundary coordinates,

$$(\hat{\alpha}^{(-v)} - \alpha^*)^\top \nabla R(\alpha^*) \;=\; \sum_{j:\, \alpha_j^*=0} \hat{\alpha}_j^{(-v)}\, \nabla R(\alpha^*)_j \;\geq\; 0,$$

and does not vanish in general. Under Assumption 2(a) and boundedness of $\mathbb{E}\Big[(\tilde{W}_i^*)^2\, \|\hat{\psi}_{\mathrm{tjs}}^{\mathrm{oof}}(X_i)\|_2^2\Big]$, the entries $\nabla R(\alpha^*)_j$ are $\mathrm{O}(1)$. The objective in (A.19) is strongly convex in α under Assumption 2(b). This effect is only reinforced by the ridge term $\lambda\|\alpha\|_2^2$, which does not affect the order of the bound. Therefore, standard M-estimation theory for strongly convex objectives under linear constraints gives $\|\hat{\alpha}^{(-v)} - \alpha^*\|_2 = \mathrm{O}_p(n^{-1/2})$, and in particular $\hat{\alpha}_j^{(-v)} = \mathrm{O}_p(n^{-1/2})$ at coordinates with $\alpha_j^* = 0$. The linear term in (A.20) is therefore $\mathrm{O}_p(n^{-1/2})$ and the quadratic term $\mathrm{O}_p(n^{-1})$, so

$$R(\hat{\psi}_{\mathrm{stack}}^{(-v)}) - R(\alpha^{*\top} \hat{\psi}_{\mathrm{tjs}}^{\mathrm{oof}}) \;=\; \mathrm{O}_p(n^{-1/2}).$$

The bound is uniform in v, and averaging over the V folds gives the corresponding statement for $\bar{R}$. This contribution is absorbed into the $\mathrm{O}_p(n^{-1/2})$ remainder of the main oracle inequality, so (13) holds for the full library $k \in \{\mathrm{tf}, \mathrm{joint}, \mathrm{sep}, \mathrm{stack}\}$.